Grounding Research in Place

An ecoregional assessment of paleofire science for restoring ecocultural landscapes

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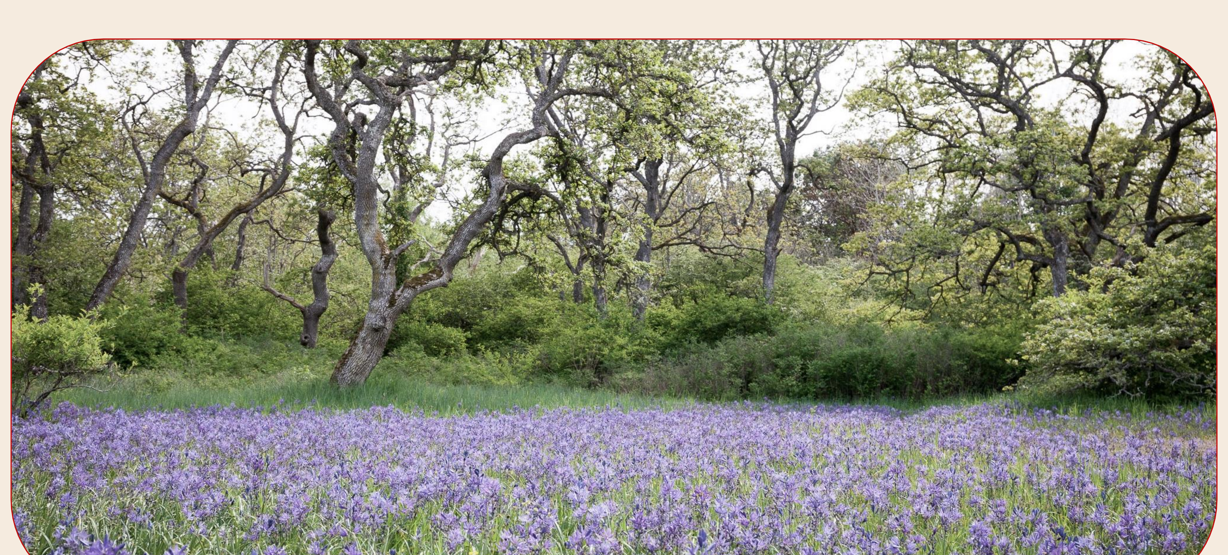
Prairie Oak Fire Regimes

The 2000s have been marked by an exponential increase in extreme wildfires in the Pacific Northwest.[1] In British Columbia, this is the product of a drier climate, land-use change, and fire suppression policies such as banning Indigenous burning in 1874.[2,3]

Prior to this, fire regimes were more complex due to the active land stewardship by Indigenous peoples using regular, low-intensity fire to maintain landscape complexity and biotic diversity to achieve a multiplicity of land management targets.[4,5]

The resulting ecologically and culturally important prairie oak landscapes of the Willamette Valley Puget Trough Georgia Basin ecoregion are now a conservation concern due to multiple compounding threats, requiring rapid restoration at scale.[6,7]

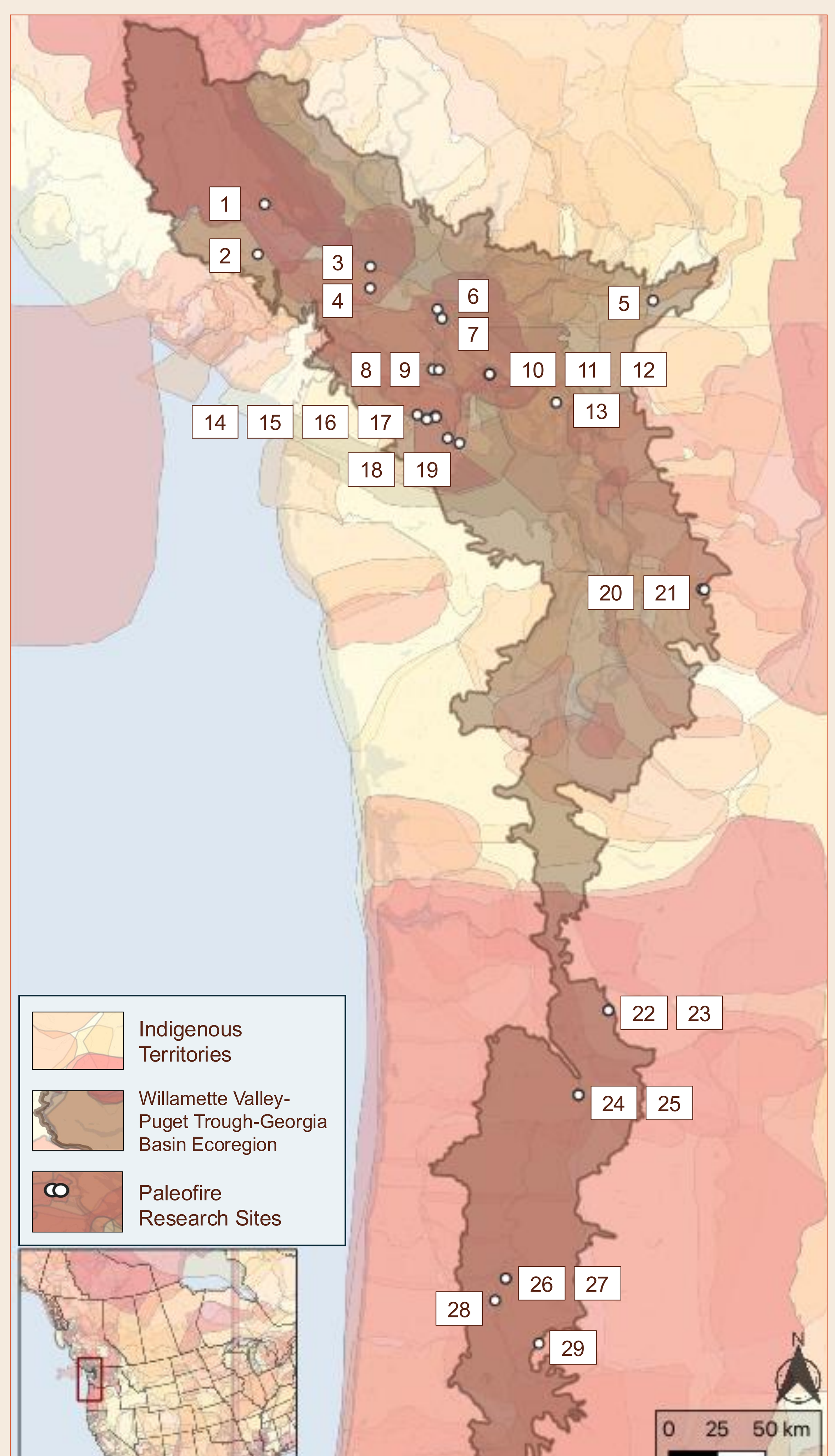


Paleoecology for Restoration

Restoration planning is transdisciplinary in that it requires ecosystem information from multiple knowledges, of which, many have identified the necessity of Indigenous fire stewardship (IFS) to ensure effective land management.[8,9] Paleoecology can provide relevant millennial scale ecosystem information, yet there is limited long-term IFS data for prairie oak ecosystems.[10]

I reviewed 29 fire history reconstruction sites to assess how data interpretation and landscape management recommendations have been influenced by methodology choices.

Fire History Reconstruction Sites in the WPG Ecoregion



Literature Review Results

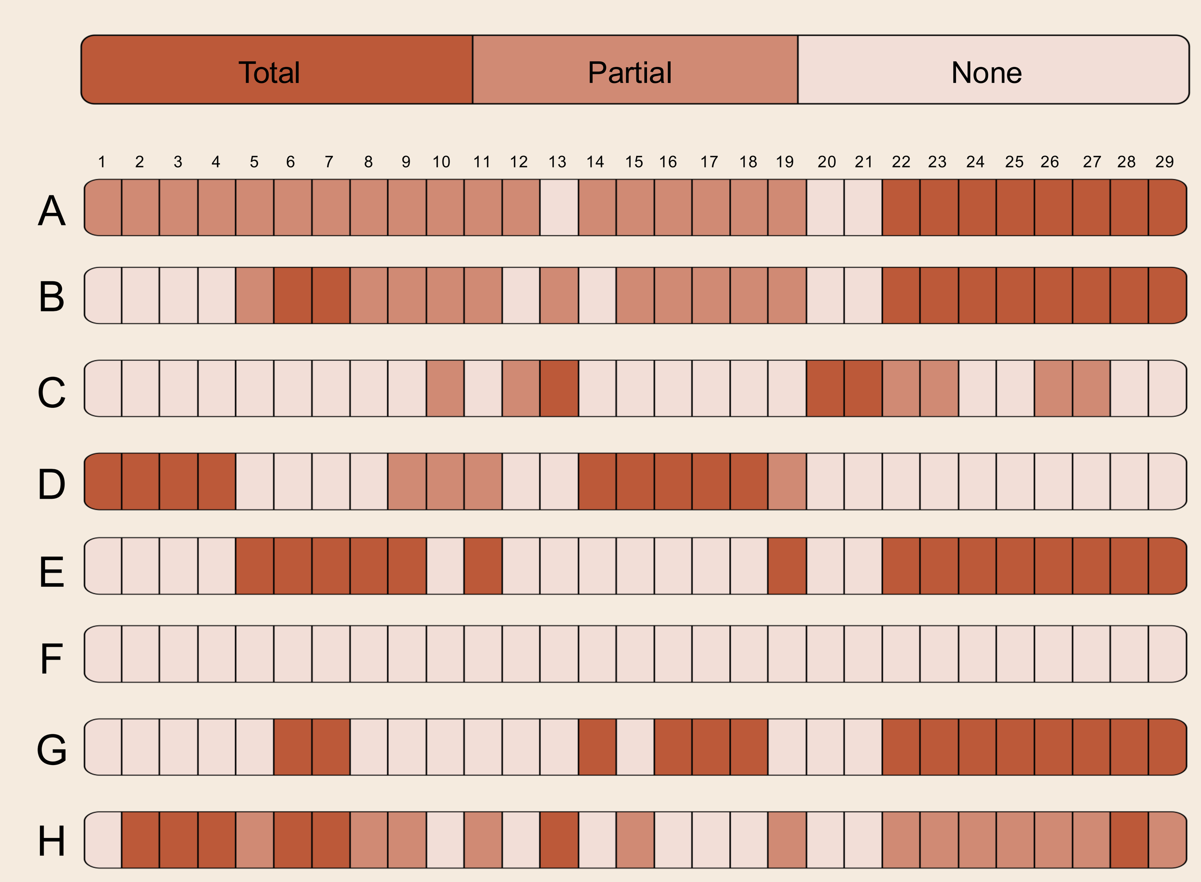
This ecoregional research gap is mirrored globally as both non-forested ecosystems and IFS have been identified as priorities for methodological advancement needed to action paleodata applications,[11,12,13,14] with recommendations to begin by evaluating:

A. Vegetation community characteristics
B. Assess potential for IFS

Then, when non-forested ecosystems and/or IFS is present, methods should involve:

C. Increased sampling frequency[15]
D. Limit charcoal chemical processing[16]
E. Decreased sieve size[17]
F. Measure charcoal features[17, 18]
G. Qualify charcoal morphology[17, 19]
H. Omit peak analysis[17, 20]

I assessed whether these recommendations have been incorporated into 29 fire history reconstruction sites within the ecoregion:



Regional ecosystems and IFS are typically considered (A, B), though most recommendations (C-H) remain unused. This gap offers the chance to adopt new methods that center IFS to better identify ecosystem dynamics and qualify the historic range of variability,[21] strengthening recommendations for restoration targets that advance static baselines toward resilience qualities.

New Research Directions

Adapting methodology to place requires additional work in challenging the prevailing significance threshold used to identify fire regime qualities, such as in reconsidering:

Shifted Baselines: The dramatic effects of colonization created pronounced signals in the fossil record that are not representative of Indigenous land management practices.[22]

Climate and People: Paleorecord signals of climate and human ecosystem controls are not easily distinguishable since humans can modulate climate patterns in many ways.[23]

Landscape Diversity: Landscape diversity paired with multiple controls on fire regimes creates spatiotemporally emergent patterns that are complex and overlapping.[12]

Multiple Sources: Rich information exists from ecology,[7] ethnography,[5] archeology,[24] historical ecology,[25] and dendrochronology[26] to aid incorporation of IFS histories into paleoresearch.[27]

By explicitly considering the unique physical features of our ecoregion and respecting the legacy, present, and future of IFS, paleoecology can embrace new research methodologies to provide data that is critical to implementing effective conservation and restoration of ecocultural landscapes.

This is a QR code for you to explore poster references, literature review details, spatial data, and the many ways to stay connected as this research continues to unfold.